

This Booklet contains 32 pages.

Do not open this Test Booklet until you are asked to do so.

Read carefully the Instructions on the Back Cover of this Test Booklet.

Important Instructions :

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on Side-1 and Side-2 carefully with blue/black ball point pen only.
2. The test is of 3 hours duration and this Test Booklet contains 180 questions. Each question carries 4 marks. For each correct response, the candidate will get 4 marks. For each incorrect response, one mark will be deducted from the total scores. The maximum marks are 720.
3. Use Blue/Black Ball Point Pen only for writing particulars on this page/markings responses.
4. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
5. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
6. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
7. Use of white fluid for correction is *not* permissible on the Answer Sheet.

Name of the Candidate (in Capitals) _____

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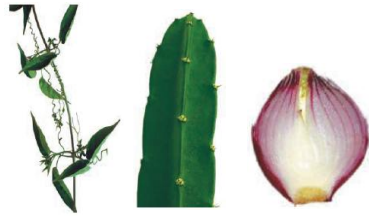
Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

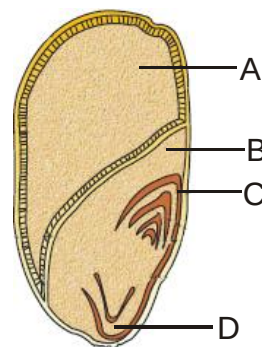
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Centre Superintendent : _____

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| <p>1. Fatty acids and glycerol are :-
 (1) Being insoluble, cannot be absorbed into lymph
 (2) Being insoluble, cannot be absorbed into blood
 (3) Being soluble, cannot be absorbed into blood
 (4) Being soluble, can be absorbed into blood</p> <p>2. Choose the incorrect option regarding a normal human :-
 (1) The skull is dicondylic
 (2) Patella is a sesamoid bone which covers the knee dorsally
 (3) Scapula is large triangular flat bone, situated on dorsal side of thorax
 (4) The pelvic girdle has two coxal bones</p> <p>3. Statements
 A- Carbonic anhydrase is present in the erythrocytes.
 B- In erythrocytes the carbondioxide combine with water and is transported.
 (1) Statement A is correct and is responsible for statement B
 (2) Statement A is not correct but Statement B is correct
 (3) Both Statement A and B are wrong
 (4) Statement A is correct and statement B is wrong</p> <p>4. Which gymnospermic plant show fungal associate in the form of mycorrhiza?
 (1) Cycas
 (2) Sequoia
 (3) Pinus
 (4) Cedrus</p> | <p>5. A sub-division of brain which is responsible for body-balance also have white matter inside:
 (1) Cerebrum
 (2) Cerebellum
 (3) Pons
 (4) Medulla oblongata</p> <p>6. Which of the following disease is correctly matched to hypersecretion of related hormone:-
 (1) Growth Hormone → Gigantism
 (2) Insulin → Diabetes Mellitus
 (3) Thyroxin → Cretinism
 (4) Aldosterone → Hypotension</p> <p>7. Which option is correct for the <i>Ficus</i> ?
 (1) It has bisexual flowers
 (2) It has only male or female flower
 (3) It has male, female and sterile flowers
 (4) It has only sterile flowers</p> <p>8. Find out the odd one (with reference to number of unit membrane) :-
 (1) Mitochondria
 (2) Nucleus
 (3) Chloroplast
 (4) Ribosome</p> <p>9. When oxygen is not available to a muscle cell, NADH formed during glycolysis passes hydrogen atom to :-
 (1) Electron transport chain
 (2) Pyruvic Acid
 (3) Acetaldehyde
 (4) Oxygen</p> |
|--|---|

- | | |
|--|---|
| <p>10. Which are exclusively marine mammals:-
 (1) Seal
 (2) Walrus
 (3) Whale
 (4) All of the above</p> <p>11. Emulsification of fats by bile take place in:-
 (1) Stomach
 (2) Large Intestine
 (3) Small Intestine
 (4) Liver</p> <p>12. What is the location of troponin in the process of muscle contraction ?
 (1) Attach to myosin filament
 (2) Attach to myosin cross bridge
 (3) Attach to Tropomyosin
 (4) Attach to T-tubules</p> <p>13. Initiation of ICBN from which year -
 (1) 1754
 (2) 1753
 (3) 1961
 (4) 1978</p> <p>14. Smallest Angiosperm and Tallest Angiosperm respectively are -
 (1) Spirullina and Chlorella
 (2) Azolla and eucalyptus
 (3) Wolfia and sequoa
 (4) Wolfia and Eucalyptus</p> <p>15. Nerve impulse travel between two neurons with the help of :
 (1) Acetylcholine and gaba
 (2) Glycine and serotonin
 (3) Glutamate and acetyl choline
 (4) Choline and acetylcholine</p> | <p>16. The major role of thymosin hormone is:-
 (1) Differentiation of T-lymphocytic
 (2) Regulate B.M.R
 (3) Regulate menstrual cycle
 (4) Elongation of Bones</p> <p>17. Identify the type of leaf modification in the given diagram (A-C) and choose correct option :-</p> <div style="text-align: center;">  <p>(A) (B) (C)</p> </div> <p>(1) A-support (spines), B-Protection (tendrill), C-Storage (fleshy leaves)
 (2) A- Support (tendrill), B-Protection (spine), C-Storage (fleshy leaves)
 (3) A-Protection (tendrill), B- Support(spine), C-storage(fleshy leaves)
 (4) A- Protection (spine), B-Support(tendrill), C-storage(fleshy leaves)</p> <p>18. If in a cell 12 tetrad are present in prophase-I then what will be the number of chromosomes in metaphase-II in each daughter cell ?
 (1) 6
 (2) 12
 (3) 24
 (4) 48</p> |
|--|---|

19. How sulfa drugs affect the bacterial pathogens ?
 (1) They dissolve the cell wall of Bacteria
 (2) They affect the ribosomes of Bacteria
 (3) They are competitive inhibitor of folic Acid synthesis in Bacteria
 (4) They stop glycolysis in the cytoplasm of a bacterial cell
20. What is common between *Columba*, *Ornithorhynchus* and *Macropus* :-
 (a) Oviparity
 (b) Homiothermy
 (c) Toothless Jaw
 (d) Functional post anal tail
 (1) a, b
 (2) b, d
 (3) b, c
 (4) a, b, c
21. Nightblindness, Cheilosis, Pellagera are respectively caused by deficiency of -
 (1) A, B₃, B₂
 (2) B₁, B₂, B₃
 (3) A, B₂, B₉
 (4) A, B₂, B₃
22. Total lung capacity is :
 (1) total volume of air accommodated in lungs at the end a forced inspiration
 (2) RV + ERV + TV + IRV
 (3) vital capacity + residual volume
 (4) All of the above
23. Whose scientists called "Darwin of 20th century"-
 (1) Ernest Haeckel
 (2) Ernest Mayer
 (3) Carolous Linnaeus
 (4) Theophrastus
24. This is not the cells of areolar connective tissue :-
 (1) Macrophages
 (2) Mast cell
 (3) Fibroblast
 (4) Schwann cells
25. What is a time-period, when a nerve is unable to conduct nerve impulse :
 (1) 1 sec. to 5 sec.
 (2) 1 minute to 5 minute
 (3) Only 1 minute
 (4) 1 milli sec. to 5 milli second.
26. M.S.H. is in related with Amphibian and reptiles :-
 (1) Metamorphosis
 (2) Metachrosis
 (3) Control sexual behaviour
 (4) Regulate blood Ca⁺² level
27. Identify A,B,C and D in given diagram:-

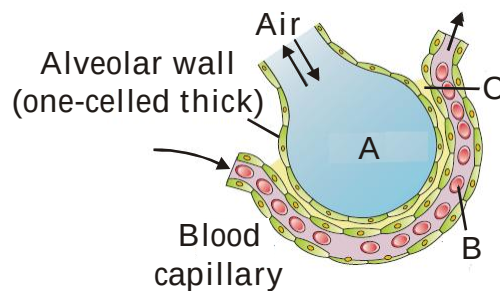


A	B	C	D
(1) Aleurone layer	Endo-sperm	Coleo-ptile	Coleo-rhiza
(2) Endo-sperm	Scute-Illum	Coleo-ptile	Coleo-rhiza
(3) Endo-sperm	Aleurone layer	Coleo-ptile	Coleo-rhiza
(4) Pericarp	Scutellum	Coleo-rhiza	Coleo-ptile

28. Which of the following is not the function of cell wall ?
 (1) Provides shape to the cell.
 (2) Protects the cell from mechanical damage and infection
 (3) Provides barrier to undesirable macromolecule
 (4) Helps in cell growth.
 (1) Only 1
 (2) Only 2
 (3) Only 1 & 2
 (4) Only 4
29. The main pressure of the cell contents against the cell wall in plant cells determined by the water content of the vacuole, is called as the :-
 (1) Water potential
 (2) Osmotic potential
 (3) DPD
 (4) Turgor pressure
30. Gambusia is a ___a___ fish belongs to class ___b___ and superclass ___c___ :-

a	b	c
(1) Parasitic	Osteichthyes	Pisces
(2) Parasitic	Chondrichthyes	Pisces
(3) Larvivorous	Osteichthyes	Pisces
(4) Larvivorous	Osteichthyes	Tetrapoda
31. Muscular sphincter which regulates the opening of oesophagus into the stomach is?
 (1) Pyloric Sphincter
 (2) Gastro-Oesophageal Sphincter
 (3) Gastric Valve
 (4) Ileocaecal Valve

32. Identify A, B and C in the given diagram and choose the correct option accordingly:



- (1) A - Alveolar cavity, B - WBC, C-Capillary wall
 (2) A-Alveolar cavity, B-RBC, C-Systemic wall
 (3) A-Alveolar cavity, B-RBC, C-Basement substance
 (4) A-Alveolar cavity, B-WBC, C-Systemic wall
33. According to R.H whittaker classification members of Monera kingdom related to-
 (1) Producer and consumer group
 (2) Decomposer and consumer group
 (3) Producer and decomposer group
 (4) Producer, decomposer and consumer
34. Non keratinized stratified squamous epithelium found in :-
 (1) Epidermis
 (2) Larynx
 (3) Fallopian tube
 (4) Pharynx and oesophagus
35. In a man, facial-nerve is injured, which one of the following will be affected ?
 (1) Swallowing
 (2) Movement of eye ball
 (3) Neck movement
 (4) Taste of anterior 2/3 part of tongue

36. A fatty acid has a carboxyl group attached to an R group. The R group could be a methyl ($-\text{CH}_3$), or ethyl ($-\text{C}_2\text{H}_5$) or higher number of $-\text{CH}_2$ groups from

- (1) 1 carbon to 19 carbon
- (2) 3 carbon to 7 carbon
- (3) 5 carbon to 6 carbon
- (4) 3 carbon to 4 carbon

37. $\oplus \text{K}_{(5)} \text{C}_{(5)} \text{A}_5 \text{G}_{(2)}$

Which option is correct for the family which has this floral formula ?

- (1) Reticulate venation, pulvinous leaf base
- (2) Bisexual flower, zygomorphic symmetry
- (3) Persistent calyx and epipetalous stamens
- (4) Berry fruit with non endospermic seed

38. The lipid molecules present in plasma membrane has polar head and non polar tail. Which of the following options represents the correct arrangement of phospholipid :-



- (1) $\begin{array}{c} \text{O} \text{ O} \text{ O} \text{ O} \\ \text{||} \text{ ||} \text{ ||} \text{ ||} \end{array}$
- (2) $\begin{array}{c} \text{O} \text{ O} \text{ O} \text{ O} \\ \text{||} \text{ ||} \text{ ||} \text{ ||} \end{array}$
- (3) $\begin{array}{c} \text{O} \text{ O} \text{ O} \text{ O} \\ \text{||} \text{ ||} \text{ ||} \text{ ||} \end{array}$
- (4) $\begin{array}{c} \text{O} \text{ O} \text{ O} \text{ O} \\ \text{||} \text{ ||} \text{ ||} \text{ ||} \end{array}$

39. The opening of stomata is not affected by:-

- (1) Oxygen concentration
- (2) Temperature
- (3) Light
- (4) Carbon dioxide concentration

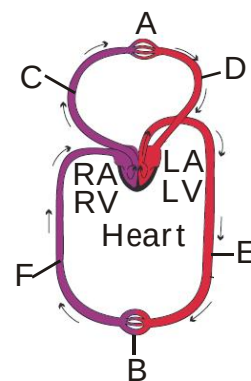
40. Animal having circular mouth, body is devoid of scales and paired fins and cranium and vertebral column are fully cartilagenous is :-

- (1) Myxine
- (2) Balanoglossus
- (3) Glossobalanus
- (4) All of the above

41. Vasa recta is Highly reduced or absent in:-

- (1) Cortical Nephrons
- (2) Juxtamedullary Nephrons
- (3) 85% Nephrons of Kidney
- (4) Both (1) and (3)

42. Identify A-F :

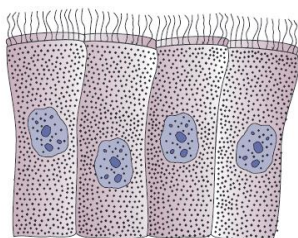


Choose the correct option

- (1) A - Lungs, B-Body parts, C-Pulmonary vein, D-Pulmonary artery, E-Dorsal aorta, F-Vena cava
- (2) A-Lungs, B-Body parts, C-Pulmonary artery, D-Pulmonary vein, E-Dorsal aorta, F- Vena Cava
- (3) A-Lungs, B-Body parts, C-Pulmonary artery, D-Pulmonary vein, E-Vena cava, F-Dorsal aorta
- (4) A-Body parts, B-Lungs, C-Pulmonary artery, D-Pulmonary vein, E-Vena cava, F-Dorsal Aorta

43. Read all statement carefully and select incorrect statement Regarding Bacteria-
- (1) Longer pili help in conjugation
 - (2) Pili do not play role in motility
 - (3) Mesosome help in cell wall formation
 - (4) Techoic acid found in gram (-)ve Bacteria

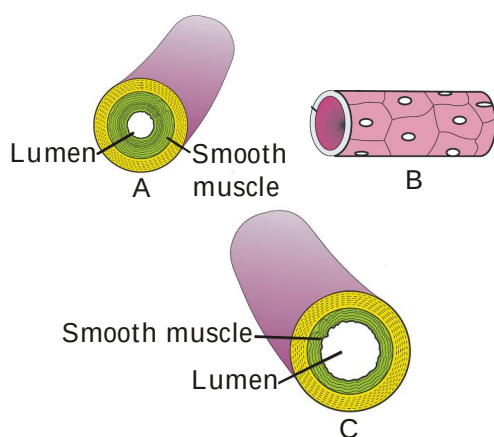
44. Following diagram represent which type of epithelium :-



- (1) Simple squamous
 - (2) Simple cuboidal
 - (3) Simple columnar
 - (4) Ciliated columnar
45. Alzheimers disease is caused by :
- (1) Lack of dopamine
 - (2) Lack of acetylcholine
 - (3) Lack of nor-adrenaline
 - (4) Lack of thyroxine
46. Which of the following is not aromatic amino acid :-
- (1) Tyrosine
 - (2) Phenylalanine
 - (3) Tryptophan
 - (4) Threonine

47. Cell sap is a :-
- (1) Living content of the cell
 - (2) Non living content of the vacuole
 - (3) Non living content of the protoplasm
 - (4) Living content of the cytoplasm
48. For which of the statement mitosis and meiosis are similar ?
- (1) For number of chromosomes in daughter cell.
 - (2) For pairing of homologous chromosome.
 - (3) Genotype of daughter cell.
 - (4) For the amount of DNA which becomes double in synthesis phase of interphase.
49. Which hormonal treatment is given to plant to stop premature leaf fall and promote the Abscission of older mature leaves ?
- (1) IAA
 - (2) GA_3
 - (3) Zeatin
 - (4) ABA
50. Which meristem regenerates leaves in monocot removed by the grazing herbivores.
- (1) Primary meristems
 - (2) Interealary meristems
 - (3) Secondary meristems
 - (4) Lateral meristem
51. Which of the following chordate have protonephridia as excretory structure?
- (1) Platyhelminthes
 - (2) Rotifers
 - (3) Amphioxus
 - (4) All of the above

52. Identify A, B and C in the given diagram



Choose the correct option :

- (1) A- Artery, B-Capillary, C- Vein
 (2) A-Artery, B-Vein, C-Capillary
 (3) A-Vein, B-Artery, C-Capillary
 (4) A-Capillary, B-Artery, C-Vein
53. Silicated cell wall found in -
 (1) Desmids & Noctiluca
 (2) Diatoms & Gonyaulax
 (3) Diatoms
 (4) Slime moulds & Euglena
54. Tendons are the structures which connect:-
 (1) Bone to bone
 (2) Muscle to bone
 (3) Muscle to muscle
 (4) Skin with muscle
55. Adaptation of eye in dark is due to :
 (1) Repletion of vision pigments in cones
 (2) Repletion of vision pigments in rods
 (3) Depletion of vision pigments in rods
 (4) Depletion of vision pigments in cones
56. Another name of Trihydroxy propane is:-
 (1) Palmitic acid
 (2) Palmitolic acid
 (3) Glycerol
 (4) Myrestic acid

57. Which of the following is an example of cell, devoid of nuclear membrane and mitochondria ?

- (1) Bacterial cell
 (2) Sperm
 (3) Protista
 (4) Sponge cell

58. Which mineral is essential for synthesis of growth hormone ?

- (1) Fe
 (2) Mn
 (3) Zn
 (4) B

59. When a brief exposure of light is given in dark duration to short day plant ? :-

- (1) Flowering response is not seen
 (2) It is converted from short day to long day plant
 (3) In that season flowering is more
 (4) Spontaneous flowering take place

60. Match the column I and column II and find out correct answer.

- | Column-I | Column-II |
|-------------------------------|---|
| (A) Bulliform cells | (I) Regulate opening and closing of stomata |
| (B) Guard cells | (II) Aerating pores in the bark of plant |
| (C) Lenticels | (III) Rolling in and out of leaves |
| (D) Cork | (IV) Phellem |
| (1) A-III ; B-I ; C-II ; D-IV | |
| (2) A-I ; B-II ; C-IV ; D-III | |
| (3) A-IV ; B-III ; C-I ; D-II | |
| (4) A-II ; B-IV ; C-III ; D-I | |

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| <p>61. Land snails are :-
 (1) Ammonotelic
 (2) Ureotelic
 (3) Excrete TMAO
 (4) Uricotelic</p> <p>62. Lymph differ from blood in possessing :-
 (1) Only WBC
 (2) More RBC & WBC
 (3) More RBC & few WBC
 (4) More WBC & few RBC</p> <p>63. Which red Algae not show red colour?
 (1) <i>Porphyra</i>
 (2) <i>Porphyridium</i>
 (3) <i>Trichodesmium</i>
 (4) <i>Batrachospermum</i></p> <p>64. In earthworm one pair male genital pores are present on :-
 (1) 14th segment mid ventral
 (2) 14th segment mid dorsal
 (3) 18th segment ventro-lateral
 (4) 18th segment dorso-lateral</p> <p>65. Which of the following structure is responsible for efficiency of transmission of sound waves to inner ear :
 (1) Helicotrema
 (2) Tympanic membrane
 (3) Ear - ossicles
 (4) Ear - pinna</p> <p>66. To perform chemical analysis of living tissue we have to grind it in :-
 (1) Monochloroacetic acid
 (2) Dichloroacetic acid
 (3) Trichloroacetic acid
 (4) Tetrachloroacetic acid</p> | <p>67. Phragmoplast grows from :-
 (1) One wall to another
 (2) Centre to wall
 (3) Wall to centre
 (4) From anywhere</p> <p>68. Name of bacteria and their function are given in column-I and column-II which one is the correct pair Identify :-</p> <table border="0" style="width: 100%; margin-left: 40px;"> <tr> <td style="text-align: center; width: 50%;">Column-I</td> <td style="text-align: center; width: 50%;">Column-II</td> </tr> <tr> <td>(A) Rhizobium</td> <td>(i) $\text{NH}_3 \rightarrow$ Nitrite ion</td> </tr> <tr> <td>(B) Nitrobacter</td> <td>(ii) Nitrate $\rightarrow \text{N}_2$ gas</td> </tr> <tr> <td>(C) Nitrosomonas</td> <td>(iii) N_2 gas $\rightarrow$ Ammonia</td> </tr> <tr> <td>(D) Pseudomonas</td> <td>(iv) Nitrite ion $\rightarrow$ Nitrate</td> </tr> </table> <p>Option-
 (1) A-(iii), B-(iv), C-(i), D-(ii)
 (2) A-(iii), B-(i), C-(iv), D-(ii)
 (3) A-(iii), B-(iv), C-(ii), D-(i)
 (4) A-(iv), B-(ii), C-(i), D-(iii)</p> <p>69. Radiolarians differs from forameniferans in being :-
 (1) Marine in nature
 (2) Belong to Rhizopoda
 (3) having a Calcareous shell
 (4) having a Silica shell</p> <p>70. The balloon-shaped structures called tyloses:-
 (1) Are extensions of xylem parenchyma cells into vessels.
 (2) Are linked to the ascent of sap through exylem vessels.
 (3) Originate in the lumen of sieve tube
 (4) Characterize the sap wood.</p> | Column-I | Column-II | (A) Rhizobium | (i) $\text{NH}_3 \rightarrow$ Nitrite ion | (B) Nitrobacter | (ii) Nitrate $\rightarrow \text{N}_2$ gas | (C) Nitrosomonas | (iii) N_2 gas $\rightarrow$ Ammonia | (D) Pseudomonas | (iv) Nitrite ion $\rightarrow$ Nitrate |
| Column-I | Column-II | | | | | | | | | | |
| (A) Rhizobium | (i) $\text{NH}_3 \rightarrow$ Nitrite ion | | | | | | | | | | |
| (B) Nitrobacter | (ii) Nitrate $\rightarrow \text{N}_2$ gas | | | | | | | | | | |
| (C) Nitrosomonas | (iii) N_2 gas $\rightarrow$ Ammonia | | | | | | | | | | |
| (D) Pseudomonas | (iv) Nitrite ion $\rightarrow$ Nitrate | | | | | | | | | | |

71. Ions excreted out through urine are mainly:-

- (1) HCO_3^- , K^+ , Ca^{+2}
- (2) Cl^- , CO_3^{2-} , HCO_3^-
- (3) K^+ , Cl^- , HCO_3^-
- (4) PO_4^{3-} , CO_3^{2-} , Mg^{2+}

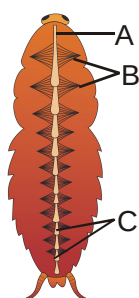
72. Micropolice man of blood :-

- (1) Neutrophil
- (2) Basophil
- (3) Eosinophil
- (4) Lymphocyte

73. Multicellular filamentous branched Rhizoids found in -

- (1) *Riccia*
- (2) *Marchantia*
- (3) *Anthoceros*
- (4) *Polytrichum*

74. In following diagram identify the structures A, B and C :-



A	B	C
(1) Alary muscles	Anterior aorta	Chamber of heart
(2) Anterior aorta	Alary muscles	Chamber of heart
(3) Chambers of heart	Alary muscles	Anterior aorta
(4) Alary muscles	Posterior aorta	Chamber of heart

75. Match the column :

Column I	Column II
(A) Cornea of eye	(i) Body - balance
(B) Cones	(ii) Peri-lymph
(C) Sacculus	(iii) Refracts light rays
(D) Scala-Tympani	(iv) Photopic vision
(1) A-iii, B-iv, C-i, D-ii	
(2) A-ii, B-iv, C-i, D-iii	
(3) A-iii, B-iv, C-iv, D-i	
(4) A-iii, B-ii, C-i, D-iv	

76. The number of thymine residues in a 5000 Kb DNA containing 23% guanine residues is (1 Kb = 1000 base)

- (1) 2.7×10^6
- (2) 2.7×10^7
- (3) 1.35×10^6
- (4) 1.35×10^7

77. During meiosis I, the number of chromosome in daughter cell :-

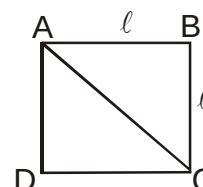
- (1) Half
- (2) Tripled
- (3) Doubled
- (4) Quadrapled

78. Which of following is not true for photo respiration?

- (1) It occurs in presence of light
- (2) Occurs in two double membrane bounded and one single membrane bounded cell organelles
- (3) Speciality of C_3 plants
- (4) Speciality of all oxygenic photosynthetic organisms

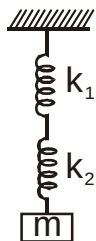
79. Following statements are correct for gemmules except :-
 (1) contains archaeocytes
 (2) covered by thin protective coat
 (3) formed during adverse condition
 (4) type of endogenous budding
80. During the secondary growth in a dicotyledonous stem, the fusiform initials of vascular cambium give rise to :-
 (1) Secondary xylem
 (2) Secondary phloem
 (3) Medullary ray
 (4) Both 1 & 2
81. Joint of skull bones are :-
 (1) Fibrous joint
 (2) Cartilaginous joint
 (3) Synovial joint
 (4) Gomphosis joint
82. The oxygen- Haemoglobin dissociation curve will show a right shift in case of :-
 (1) Less H^+ concentration
 (2) High partial pressure of O_2
 (3) Low partial pressure of CO_2
 (4) High 2,3,D.P.G.
83. In given following examples How many are ferns?
Selaginella, Marselia, Azolla, Psilotum, Lycopodium, Adiantum, Equisetum, Pteridium, Pteris
 (1) Four
 (2) Five
 (3) Six
 (4) Seven
84. Frog respire in water by :-
 (1) Lungs
 (2) Buccopharyngeal cavity
 (3) Gills
 (4) Skin
85. Which of the following function is not related with insulin hormone :-
 (1) Stimulate Glycogenesis
 (2) Inhibit Glycogenolysis
 (3) Inhibit Gluconeogenesis
 (4) Inhibit Protein synthesis
86. Match the column :-
- | Column-I | Column-II |
|---------------|-----------------------------------|
| i. Rose | p. Apocarpous
multi carpellary |
| ii. Pea | q. Diadelphous
stamens |
| iii. Hibiscus | r. Polyadelphous
stamens |
| iv. Lemon | s. Syncarpous
multicarpellary |
- (1) i-q,ii-r,iii-p,iv-s
 (2) i-p,ii-q,iii-r,iv-s
 (3) i-q,ii-r,iii-s,iv-p
 (4) i-p,ii-q,iii-s,iv-r
87. Which is true for Golgi apparatus ?
 (1) Is found only in animals.
 (2) Is found in prokaryotes.
 (3) Is a site of rapid ATP production.
 (4) Modifies and packages protein.

88. After removal of thylakoid, If we put leaf in CO_2 and H_2O containing medium in presence of sunlight, hexose is not the final product, because..... :-
- (1) Molecules which obtain light are absent
 - (2) Molecules which accept CO_2 from atmosphere are absent
 - (3) Enzymes which fixes CO_2 are absent
 - (4) CO_2 fixation does not occur in presence of light
89. Platyhelminthes are mostly parasite with dorso-ventrally flattened body. Tegument present in platyhelmenthes indicates that:-
- (1) It is thick cuticle
 - (2) Secreted by epidermis
 - (3) Protect parasite against digestive juice
 - (4) All of the above
90. As compared to spring wood, autumn wood has :-
- (1) More number of xylary elements with wider vessels
 - (2) More number of xylary elements with narrow vessels.
 - (3) Fewer xylary elements with wider vessels
 - (4) Fewer xylary elements with narrow vessels
91. The component of $\vec{A} = \hat{i} + \hat{j} + 5\hat{k}$ perpendicular to $\vec{B} = 3\hat{i} + 4\hat{j}$ is
- (1) $-\frac{4}{25}\hat{i} + \frac{3}{25}\hat{j} + 5\hat{k}$
 - (2) $-\frac{8}{25}\hat{i} - \frac{6}{25}\hat{j} + 5\hat{k}$
 - (3) $\frac{4}{25}\hat{i} - \frac{3}{25}\hat{j} + 5\hat{k}$
 - (4) $+\frac{8}{25}\hat{i} - \frac{6}{25}\hat{j} + 5\hat{k}$
92. A particle moves under the effect of a force $F = Cx$ from $x = 0$ to $x = x_1$. The work done in the process is
- (1) Cx_1^2
 - (2) $\frac{1}{2}Cx_1^2$
 - (3) Cx_1
 - (4) Zero
93. Figure square a thin square plate ABCD mass of the plate is M. The moment of inertia of the plate about side AC is :-



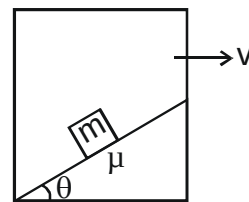
- (1) $\frac{MI^2}{18}$
- (2) $\frac{MI^2}{12}$
- (3) $\frac{MI^2}{6}$
- (4) $\frac{MI^2}{4}$

94. A mass is suspended separately by two different springs in successive order then time periods is t_1 and t_2 respectively. If it is connected by both spring as show in figure then time period is t_0 , the correct relation is:-

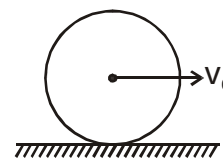


- (1) $t_0^2 = t_1^2 + t_2^2$
 (2) $t_0^{-2} = t_1^{-2} + t_2^{-2}$
 (3) $t_0^{-1} = t_1^{-1} + t_2^{-1}$
 (4) $t_0 = t_1 + t_2$
95. A tuning fork arrangement (pair) produces 4 beats/second with one fork of frequency 288 vibration/s. A little wax is placed on the unknown fork and it then produces 2 beats/sec. The frequency of the unknown fork is :-
- (1) 286 vib/s
 (2) 292 vib/s
 (3) 294 vib/s
 (4) 288 vib/s
96. In a system of units if force (F), acceleration (A) and time (T) are taken as fundamental units, then the dimensional formula of energy is :-
- (1) $[FA^2T]$
 (2) $[FAT^2]$
 (3) $[FA^2T^3]$
 (4) $[FAT]$

97. A block of mass 'm' is placed on a rough inclined plane. The plane is moving with a constant velocity v . The block is not sliding on the inclined plane. Work done by force of friction (w.r.t. ground) in time t is :-



- (1) $(\mu mg \cos \theta)vt$
 (2) zero
 (3) $(mgvt) \sin \theta \cos \theta$
 (4) $(mg \sin \theta)vt$
98. A disc of mass m is thrown in horizontal direction on a rough horizontal surface with a speed v_0 . The speed of the disc when it starts pure rolling is:-



- (1) $\frac{v_0}{2}$
 (2) $\frac{v_0}{3}$
 (3) $\frac{2v_0}{3}$
 (4) $\frac{3v_0}{3}$

99. A particle is executing simple harmonic motion with frequency f . The frequency at which its kinetic energy change into potential energy is:-

- (1) $f/2$
- (2) f
- (3) $2f$
- (4) $4f$

100. A strip is made by joining two strips of iron (α_1) and copper ($\alpha_2 > \alpha_1$), when it is heated?

- (1) Its length does not change
- (2) It gets twisted
- (3) It bends with iron on concave side
- (4) It bends with iron on convex side

101. The equation of wave is given by

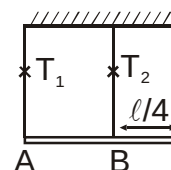
$y = A \sin \omega \left[\frac{x}{v} - k \right]$, where ω is the angular velocity and v is the linear velocity. Write dimensional formula of k .

- (1) $[M^0 L^0 T^1]$
- (2) $[M^1 L^0 T^{-1}]$
- (3) $[M^0 L^1 T^1]$
- (4) $M^1 L^{-1} T^1$

102. A long spring is stretched by 2 cm, its potential energy is U . If the spring is stretched by 1mm, the potential energy stored in it will be

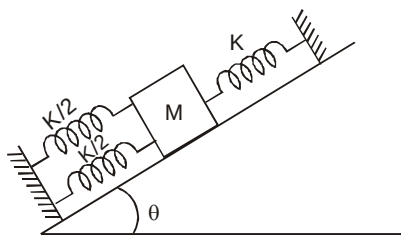
- (1) $\frac{U}{400}$
- (2) $\frac{U}{5}$
- (3) $5U$
- (4) $25U$

103. A uniform rod of length l and mass m is hung from two strings of equal length from a ceiling as shown. Tension in the two strings is :-



- (1) $T_1 = \frac{Mg}{3}, T_2 = \frac{2Mg}{3}$
- (2) $T_1 = T_2 = \frac{Mg}{2}$
- (3) $T_1 = \frac{2Mg}{3}, T_2 = \frac{Mg}{3}$
- (4) $T_1 = Mg, T_2 = 0$

104. On a smooth inclined plane, a body of mass M is attached between springs as shown. The other ends of the springs are fixed to firm supports. If each spring has force constant K , the period of oscillation of the body is :- (assuming the springs as massless)

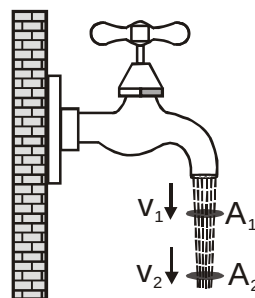


- (1) $2\pi \left(\frac{M}{2K} \right)^{1/2}$
 (2) $2\pi \left(\frac{2M}{K} \right)^{1/2}$
 (3) $2\pi \frac{Mg \sin \theta}{2K}$
 (4) $2\pi \left(\frac{2Mg}{K} \right)^{1/2}$
105. Liquids A and B are at 30°C and 20°C . When mixed in equal masses, the temperature of the mixture is found to be 26°C . Their specific heats are in the ratio of :-
 (1) 3 : 2
 (2) 1 : 1
 (3) 2 : 3
 (4) 4 : 3
106. A ball is projected from the ground at angle θ with the horizontal. After 1 second it is moving at angle 45° with the horizontal and after 2 seconds it is moving horizontally. The speed of projection of the ball is :
 (1) $10\sqrt{3} \text{ ms}^{-1}$
 (2) $20\sqrt{3} \text{ ms}^{-1}$
 (3) $10\sqrt{5} \text{ ms}^{-1}$
 (4) $20\sqrt{2} \text{ ms}^{-1}$

107. A body of mass m kg is rotating in a vertical circle is attached at the end of a string of length r metre. The magnitude of difference in the kinetic energy at the top and the bottom of the circle is :

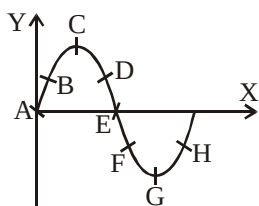
- (1) $\frac{mg}{r}$
 (2) $\frac{2mg}{r}$
 (3) $2mgr$
 (4) mgr

108. Water falling vertically from the mouth of a tap in streamline flow forms a tapering column as shown in the figure. Which of the following is the most correct explanation for this ?



- (1) As the water falls down, its velocity increases resulting in decrease in pressure. So, it is then compressed by the surrounding air
 (2) The mass of water passing through any cross-section must remain constant. So in accordance with $AV = \text{constant}$ (as liquid is incompressible), the area of cross-section decreases with increase in velocity
 (3) Due to surface tension effect, exposed surface area of liquid decreases
 (4) To attain terminal velocity, area of cross-section decreases

109. A transverse wave is travelling along a string from left to right. The figure given below represents the shape of the string at a given instant. At this instant :-



- (1) Points D, E, F have positive velocity upwards.
 (2) Points, A, B and H have negative velocity downwards.
 (3) Points C and G have zero velocity
 (4) All of these

110. An ideal gas is expanding such that $PT^2 = \text{constant}$. The coefficient of volume expansion of the gas is :-

- (1) $\frac{1}{T}$
 (2) $\frac{2}{T}$
 (3) $\frac{3}{T}$
 (4) $\frac{4}{T}$

111. A projectile is fired at 30° with momentum p , neglecting friction the change in kinetic energy when it returns to the ground will be?

- (1) zero (2) 30%
 (3) 60% (4) 100%

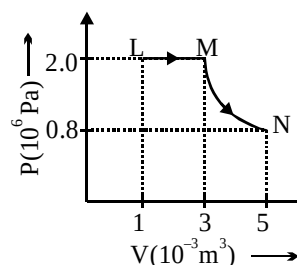
112. A motor cyclist going round in a circular track at constant speed has

- (1) Constant linear velocity
 (2) Constant acceleration
 (3) Constant angular velocity
 (4) Constant force

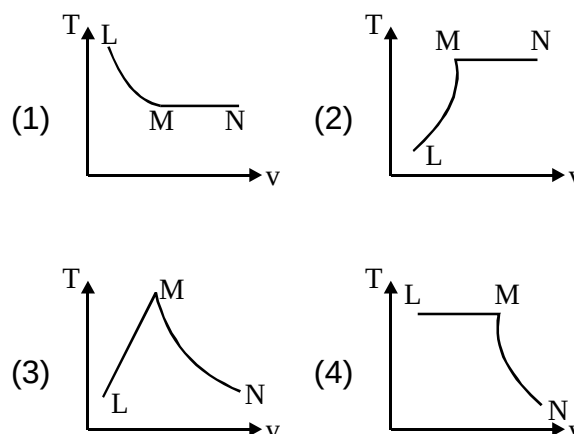
113. Water rises in a capillary tube to a certain height such that the upward force due to surface tension is balanced by 75×10^{-4} N, force due to the weight of the liquid. If the surface tension of water 6×10^{-2} N/m, the inner circumference of the capillary must be :

- (1) 1.25×10^{-2} m
 (2) 0.50×10^{-2} m
 (3) 6.5×10^{-2} m
 (4) 12.5×10^{-2} m

114. A fixed mass of ideal gas undergoes changes of pressure and volume starting at L, as shown in figure.



Which graph shows how temperature (measured in Kelvin) changes with volume ?



115. A lead bullet strikes against a steel plate with a velocity 200 ms^{-1} . If the impact is perfectly inelastic and the heat produced is equally shared between the bullet and the target, then the rise in temperature of the bullet is :-

(specific heat capacity of lead = $125 \text{ J kg}^{-1} \text{ K}^{-1}$)

- (1) 80°C
- (2) 60°C
- (3) 160°C
- (4) 40°C

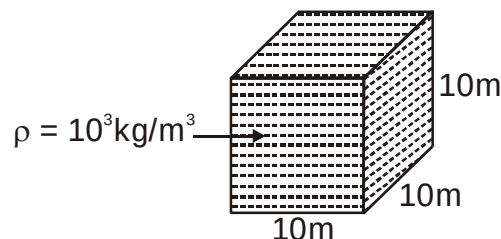
116. A particle moves in the xy-plane with velocity $u_x = 8t - 2$ and $u_y = 2$. If it passes through the point (14, 4) at $t = 2 \text{ s}$, the equation of its path is :

- (1) $x^2 = 2 - y$
- (2) $x = 2 - y^2$
- (3) $x = y^2 - y + 2$
- (4) $x = y^2 + y - 2$

117. A bridge is convex with radius of curvature r . A car passing through it with speed v , then normal reaction by the road at the lowest point will be:

- (1) $mg + \frac{mv^2}{r}$
- (2) $mg - \frac{mv^2}{r}$
- (3) $\frac{m^2v^2g}{r}$
- (4) $\frac{v^2g}{r}$

118. If atmospheric pressure is neglected, ratio of force at the base to the one side walls of cubical container will be :-



- (1) 2
- (2) 1
- (3) $1/2$
- (4) None

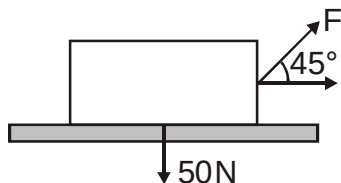
119. Two identical spheres of same material are in contact with each other. If r be the radius of the sphere, then gravitational attraction between them is proportional to:-

- (1) r
- (2) r^2
- (3) r^3
- (4) r^4

120. In a pressure cooker, the cooking is fast, because-

- (1) The boiling point of water is raised by the increased pressure inside the cooker
- (2) The boiling point of water is lowered by pressure
- (3) More steam is available to cook the food at 100°C
- (4) More pressure is available to cook the food at 100°C

121. A body of weight 50N is dragged on a horizontal surface with a force of 28.2 N. If the body remains stationary, then the frictional force acting on the body and the normal reaction force will be :-

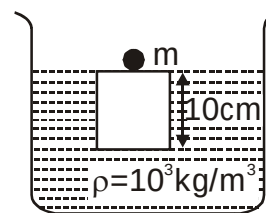


- (1) 10N, 15N
(2) 5N, 6N
(3) 2N, 3N
(4) 20N, 30N
122. A body is projected vertically upward with a velocity of 10 ms^{-1} . It reaches the maximum vertical height h in time t . In

$\frac{t}{2}$, the height covered is:-

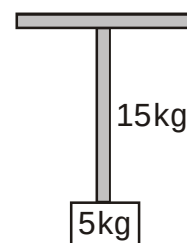
- (1) $\frac{h}{2}$
(2) $\frac{2h}{5}$
(3) $\frac{3h}{4}$
(4) $\frac{5h}{8}$

123. In the given figure, if mass m is removed from the top of cubical block. Then block comes up by 3cm, then value of m will be:-



- (1) 200 gm
(2) 300 gm
(3) 400 gm
(4) None

124. A uniform rope of length 10 m and mass 15 kg hangs vertically from a rigid support. A block of mass 5 kg is attached to the free end of the rope. A transverse pulse of wavelength 0.08 m is produced at the lower end of the rope. The wavelength of the pulse when it reaches the top of the rope will be :-



- (1) 0.08 m
(2) 0.04 m
(3) 0.16 m
(4) 0 m

125. Two liquids A and B are at temperature of 100°C and 20°C . Their masses are in the ratio of 2:3 and their specific heats in the ratio 3:4. If these two liquids are mixed then the resulting temperature will be ?

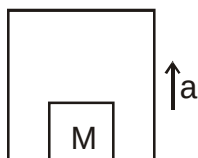
(1) $\frac{160}{3}^{\circ}\text{C}$

(2) $\frac{140}{3}^{\circ}\text{C}$

(3) $\frac{130}{3}^{\circ}\text{C}$

(4) $\frac{110}{3}^{\circ}\text{C}$

126. With what acceleration should the box of figure moving up so that the block of mass M exerts a force $3Mg/2$ on the floor of the box ?



(1) $\frac{g}{4}$

(2) $\frac{g}{2}$

(3) $\frac{3g}{4}$

(4) $4g$

127. Two particles of masses 1 kg and 2 kg are located at points (1, 2) and (-1, 3) respectively. The coordinates of COM are:-

(1) $(1/3, 8/3, 0)$

(2) $(0, 1/3, 8/3)$

(3) $(-1/3, 8/3, 0)$

(4) $(8/3, -1/3, 0)$

128. If the terminal speed of a sphere of gold (density = 19.5 gm/cc) is 0.2 m/s in a viscous liquid (density = 1.5 gm/cc), find the terminal speed of a sphere of silver (density = 10.5 gm/cc) of the same size in the same liquid.

(1) 0.1 m/s

(2) 0.2 m/s

(3) 0.4 m/s

(4) 0.133 m/s

129. A source of sound is travelling towards a stationary observer. The frequency of sound heard by the observer is three times the original frequency. The velocity of sound is $V\text{ m/s}$. The speed of source will be:-

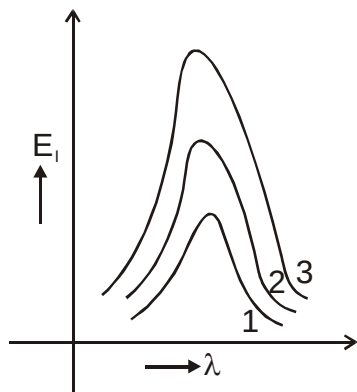
(1) $\frac{2}{3}V$

(2) V

(3) $\frac{3}{2}V$

(4) $3V$

130. The graph shown in the figure represents energy density E_λ versus λ for three sources sun, welding arc and tungsten filament. For $\lambda_{\max}$, correct combination will be :-



- (1) 1–tungsten, 2–welding arc, 3–sun
 (2) 1–sun, 2–tungsten, 3–welding arc
 (3) 1–sun, 2–welding arc, 3–tungsten
 (4) 1–welding arc, 2–sun, 3–tungsten
131. A cricket ball of mass 500 g collides with a bat with velocity 15 m/s and returns with the same velocity within 0.01 second. The force acted on bat is:-
 (1) Zero
 (2) 750 N
 (3) 1500 N
 (4) 2000 N
132. Two boys A and B of masses 55 kg and 65 kg are sitting at the two ends of a boat which is at rest in still water. The boat weighs 100 kg and is 3.0 m long. Boy A walks towards B and shakes hand. The displacement of boat is :-
 (1) 0
 (2) 0.75m
 (3) 3.0m
 (4) 2.3m

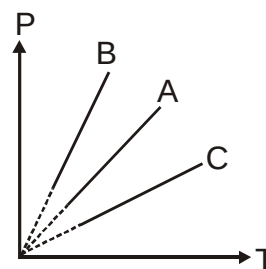
133. The total energy of a particle executing S.H.M is proportional to :-

- (1) Displacement from equilibrium position
 (2) Frequency of oscillation
 (3) Velocity in equilibrium position
 (4) Square of amplitude of motion

134. In a transverse progressive wave of amplitude A , the maximum particle velocity is eight times its wave velocity. The wavelength of the wave is :-

- (1) $\frac{\pi A}{4}$
 (2) $\frac{\pi A}{2}$
 (3) πA
 (4) $2\pi A$

135. Pressure versus temperature graph of the ideal gas at constant volume V is shown by a straight line A. Now, mass of the gas is doubled and the volume is halved; then the corresponding pressure versus temperature graph will be shown by the line :-



- (1) A
 (2) B
 (3) C
 (4) none of these

136. The reactions of heavy water are slow. The reason is:-

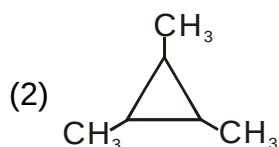
- (1) Heavy water is associated
- (2) Heavy water is dissociated
- (3) High bond energy of D-O bond
- (4) Heavy water is of lower mass

137. Which is the right structure of XeF_4 :-

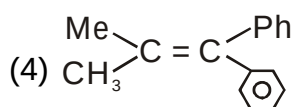
- (1)
- (2)
- (3)
- (4)

138. Which of the following can not show G.I. :-

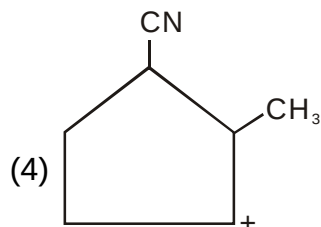
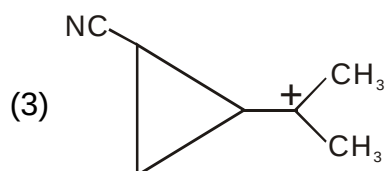
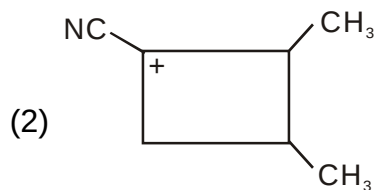
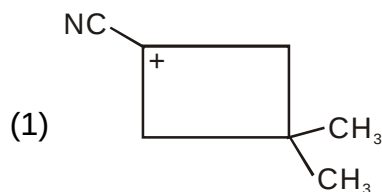
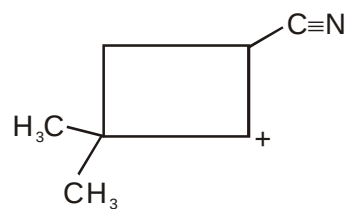
- (1) $\text{Cl}-\text{CH}=\text{CH}-\text{Ph}$



- (3) $\text{H}_3\text{C}-\text{CH}=\text{C}=\text{C}=\text{CH}-\text{CH}_3$



139. The following carbocation rearranges to give :-



140. Which of the following is correct option for free expansion of an ideal gas under adiabatic condition?

- (1) $q = 0, \Delta T < 0, W \neq 0$
- (2) $q = 0, \Delta T \neq 0, W = 0$
- (3) $q \neq 0, \Delta T = 0, W = 0$
- (4) $q = 0, \Delta T = 0, W = 0$

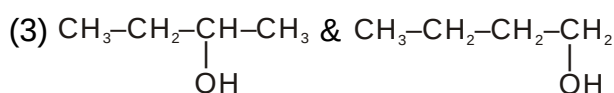
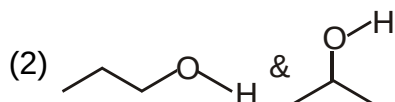
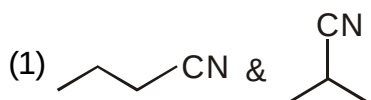
141. 3 centered 2 electron bond is present in :-

- (1) Be_2H_4 (2) $\text{Al}_2(\text{CH}_3)_6$
(3) B_2H_6 (4) All of these

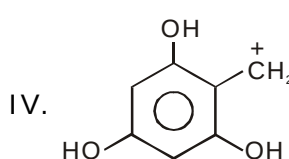
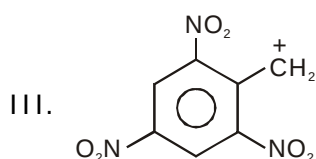
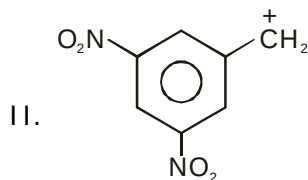
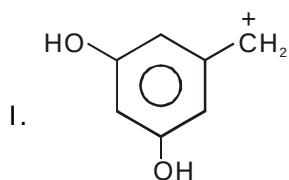
142. In which of the following N is in the sp^2 hybridisation state :-

- (1) $(\text{CH}_3)_3\text{N}$ (2) NO_2
(3) CH_3CN (4) NO_2^+

143. Which of the following is not position isomers:-



144. Arrange the following carbocations in decreasing order of stability (most stable first).



- (1) $\text{IV} > \text{I} > \text{II} > \text{III}$
(2) $\text{III} > \text{IV} > \text{I} > \text{II}$
(3) $\text{II} > \text{III} > \text{I} > \text{IV}$
(4) $\text{II} > \text{I} > \text{III} > \text{IV}$

145. The energy of electron in the 2^{nd} orbit of Li^{+2} is :-

- (1) -10.2 eV (2) 10.2 eV
(3) -30.6 eV (4) 30.6 eV

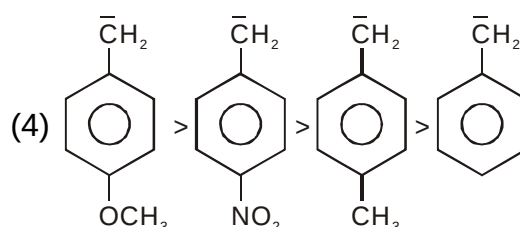
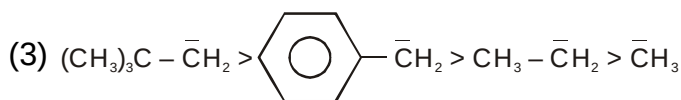
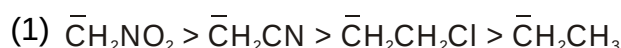
146. The hybridization of N-atom and shape of $(\text{SiH}_3)\text{NCO}$ around N is :-

- (1) sp^2 , planar
(2) sp^3 , planar
(3) sp^3 , pyramidal
(4) sp , linear

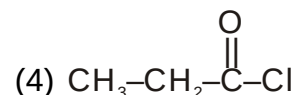
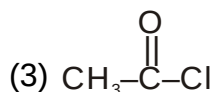
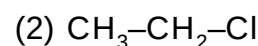
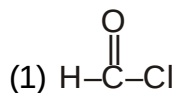
147. The element with atomic number 35 belongs to :-

- (1) d-Block (2) f-Block
(3) p-Block (4) s-Block

148. Which of the following order(s) correctly depict the decreasing order of stability of carbanion?



149. Ethanoyl chloride is the IUPAC name of :-



150. If the de-Broglie wavelength of the third Bohr orbit of hydrogen atom is $8\text{\AA}$, the circumference of the orbit will be:-

- (1) $4\text{\AA}$ (2) $8\text{\AA}$
(3) $24\text{\AA}$ (4) $16\text{\AA}$

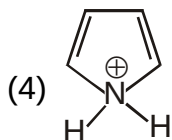
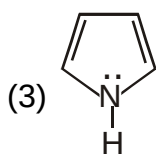
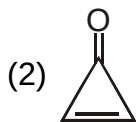
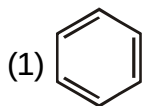
151. Which of the following give methane gas on hydrolysis :-

- (1) Al_4C_3 (2) Mg_2C_3
(3) CaC_2 (4) None

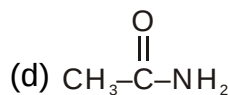
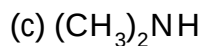
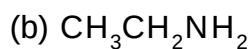
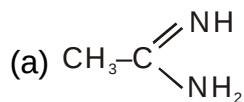
152. The correct order of size would be :-

- (1) $\text{Ni} < \text{Pd} = \text{Pt}$
(2) $\text{Pd} < \text{Pt} < \text{Ni}$
(3) $\text{Pt} < \text{Ni} < \text{Pd}$
(4) $\text{Pd} > \text{Pt} > \text{Ni}$

153. Which of the following is not Aromatic:-



154. The correct order of basicities of the following compounds is :-



- (1) $b > a > c > d$
(2) $a > c > b > d$
(3) $c > a > b > d$
(4) $a > b > c > d$

155. A sample of PCl_3 contains 1.4 moles of the substance. How many atoms are there in the sample ?

- (1) 4
(2) 5.6
(3) 8.431×10^{23}
(4) 3.372×10^{24}

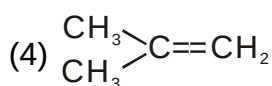
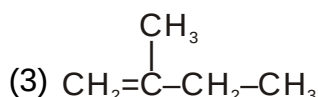
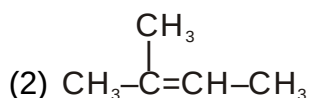
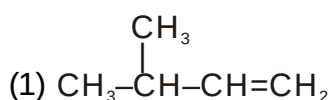
156. Match the columns :

Column-I	Column-II
(p) Pb_3O_4	(i) Neutral oxide
(q) NO	(ii) Acidic oxide
(r) CrO_3	(iii) Basic oxide
(s) K_2O	(iv) Mixed oxide
(1) p-iii, q-ii, r-iv, s-i	
(2) p-iv, q-i, r-ii, s-iii	
(3) p-iv, q-i, r-iii, s-ii	
(4) p-iii, q-ii, r-i, s-iv	

157. The ionisation energy will be maximum for the process :-

- (1) $\text{Ba} \rightarrow \text{Ba}^{+2}$
- (2) $\text{Be} \rightarrow \text{Be}^{+2}$
- (3) $\text{Cs} \rightarrow \text{Cs}^{+}$
- (4) $\text{Li} \rightarrow \text{Li}^{+}$

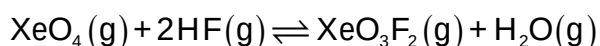
158. Which of the following alkene has highest value of HOC :-



159. If K_1 and K_2 are the respective equilibrium constants for the two reactions :-



The equilibrium constant for the reaction



will be

- (1) K_2/K_1
- (2) $K_1 \cdot K_2$
- (3) K_1/K_2
- (4) $K_1 \cdot K_2 \cdot K_3$

160. Number of atoms in 24 g of He is :-

- (1) N_A
- (2) $2 N_A$
- (3) $4 N_A$
- (4) $6 N_A$

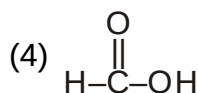
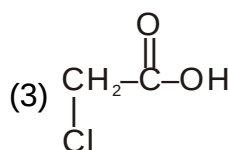
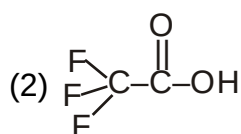
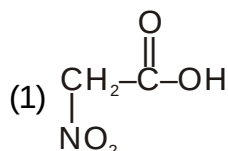
161. Select the correct statement :

- (1) LiHCO_3 can exist in both solution phase and solid state
- (2) Sodium hypochlorite (NaClO) contains O^{2-} ions
- (3) Both LiOH and NaOH are strong water soluble bases
- (4) Thermal stability of superoxides of alkali metals increases down the group.

162. The pair of amphoteric hydroxide is :-

- (1) $\text{Al}(\text{OH})_3$, LiOH
- (2) $\text{Be}(\text{OH})_2$, $\text{Mg}(\text{OH})_2$
- (3) $\text{B}(\text{OH})_3$, $\text{Be}(\text{OH})_2$
- (4) $\text{Be}(\text{OH})_2$, $\text{Zn}(\text{OH})_2$

163. Which of the following is most acidic :-



164. For a reaction, $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$, the value of K_c does not depend on :-

- (1) Initial concentration of reactants
- (2) Temperature
- (3) Pressure
- (4) 1, 3 both

165. When a real gas shows behaviour same as ideal gas :-

- (1) At low temperature and low pressure
- (2) At high temperature and high pressure
- (3) At low temperature and high pressure
- (4) At high temperature and low pressure

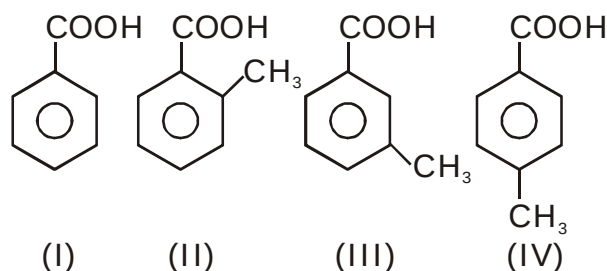
166. The octet rule is not obeyed in :-

- (1) CO_2
- (2) BCl_3
- (3) PCl_5
- (4) (2) and (3) both

167. The Z_{eff} order is :-

- (1) $\text{Fe} < \text{Fe}^{+2} < \text{Fe}^{+3}$
- (2) $\text{Fe}^{+3} < \text{Fe}^{+2} < \text{Fe}$
- (3) $\text{Fe}^{+2} < \text{Fe} < \text{Fe}^{+3}$
- (4) $\text{Fe}^{+2} < \text{Fe}^{+3} < \text{Fe}$

168. Arrange the following in correct order of acidic strength :-



- (1) $\text{I} > \text{II} > \text{III} > \text{IV}$
- (2) $\text{I} > \text{IV} > \text{III} > \text{II}$
- (3) $\text{II} > \text{I} > \text{III} > \text{IV}$
- (4) $\text{II} > \text{I} > \text{IV} > \text{III}$

169. The solubility of AgBr will be minimum in:-

- (1) pure water
- (2) 0.2 M NaCl
- (3) 0.1 M NH_3
- (4) 0.2 M NaBr

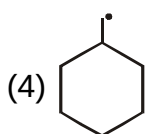
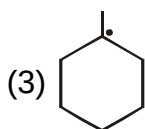
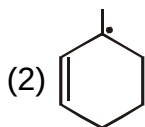
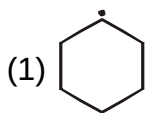
170. Which gas will have maximum root mean square velocity at same temperature :

- (1) He
- (2) N_2
- (3) O_2
- (4) SO_2

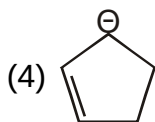
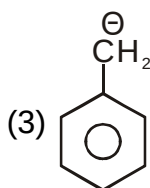
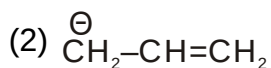
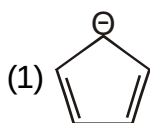
171. Which of the following compounds does not contain coordinate bond :-

- (1) O_3
- (2) H_2O
- (3) HNO_3
- (4) N_2O_5

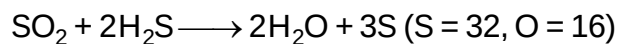
172. Most stable radical among following :-



173. Which of the carbanion is most stable :-



174. 1 mol of SO_2 and 1 mol of H_2S react completely to form



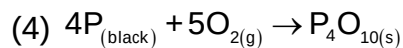
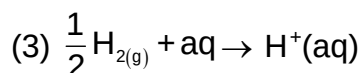
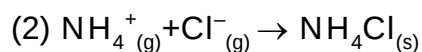
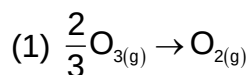
(1) 96 gm S

(2) 48 gm S

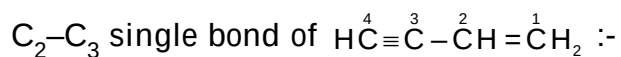
(3) 24 gm S

(4) 64 gm S

175. Which of the following represents heat of formation of the product:-



176. The hybridization of carbon atoms in



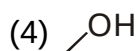
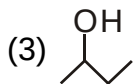
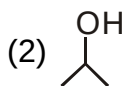
(1) sp^3-sp^3

(2) sp^2-sp

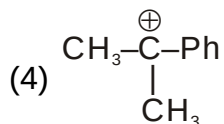
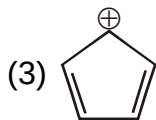
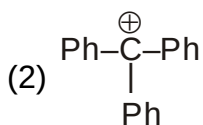
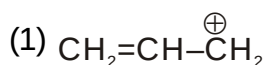
(3) $\text{sp}-\text{sp}^2$

(4) sp^3-sp

177. What is the final product when propene reacts with dil. H_2SO_4 :



178. Which of the following is most stable carbocation :-



179. Which of the following would have the largest buffer capacity towards addition of small amounts of acid or base ?

(Benzoic acid is a weak acid)

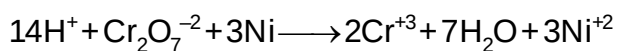
(1) A solution 0.80 M in benzoic acid and 0.10M in sodium benzoate

(2) A solution 0.80 M in benzoic acid and 0.80M in sodium benzoate

(3) A solution 0.80 M in HCl and 0.80 M in KCl

(4) A solution 0.80 M in benzoic acid and 0.40M in sodium benzoate

180. Which species is acting as a reducing agent in the following reaction.



(1) $\text{Cr}_2\text{O}_7^{2-}$

(2) Ni

(3) H^+

(4) H_2O

SPACE FOR ROUGH WORK

SPACE FOR ROUGH WORK

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1. Each candidate must show on demand his/her Allen ID Card to the Invigilator.
2. No candidate, without special permission of the Invigilator, would leave his/her seat.
3. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty.
4. Use of Electronic/Manual Calculator is prohibited.
5. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Hall. All cases of unfair means will be dealt with as per Rules and Regulations of this examination.
6. No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.
7. The candidates will write the Correct Name and Form No. in the Test Booklet/Answer Sheet.

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